

XAI Lecture 24 — Explaining LLM Reasoning

Background, then two papers: generate the reasoning (Rajani 2019) and localise the evidence (Yin & Neubig 2022)

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

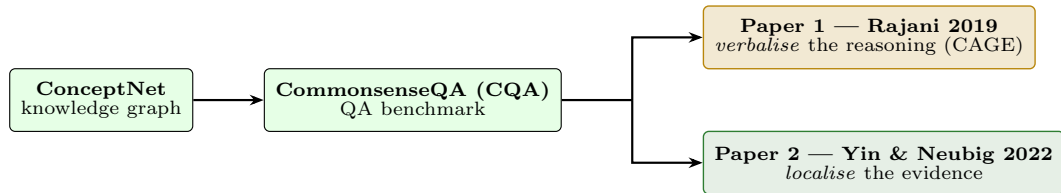
Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>



Background - the problem before the two papers ---

Both papers attack the same question — **how do large language models (LLMs, or LMs) reason?** — from opposite ends. Before that, three things must be on the table:



- **What commonsense reasoning is**, and why it is hard for NLU (Natural Language Understanding) systems.
- **ConceptNet** — the knowledge graph (Speer et al., 2017).
- **CommonsenseQA (CQA)** — the question-answering (QA) benchmark built *from* ConceptNet (Talmor et al., 2019); the lecture's shared backdrop, and the benchmark **Paper 1** builds on.

🍷 Why commonsense is hard for NLU ---

When people answer, they draw on world knowledge **not present in the text**: space, cause and effect, social conventions.

Example (Talmor et al., 2019):

“Where was Simon when he heard the lawn mower?”

A human silently infers: a lawn mower is **outdoors**, at **street level** → Simon was **outside**.

The contrast

Classic reading-comprehension QA — e.g. **SQuAD** (*Stanford Question Answering Dataset*, Rajpurkar et al. 2016: questions whose answer is a *span of text* inside a given paragraph) — hands you that paragraph and asks about *it*. Commonsense QA gives **no** paragraph: the knowledge must already be in the model.

Remarks

“Trivial for humans, out of reach for NLU systems” — the gap this whole lecture circles around.

🍷 ConceptNet: a commonsense knowledge graph (Speer et al., 2017) ---

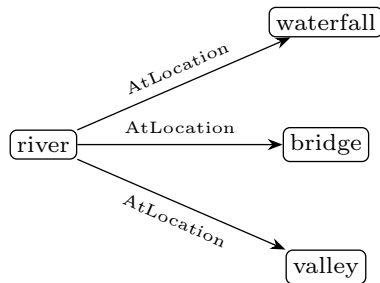
ConceptNet stores everyday knowledge as a **graph of triples**:

- nodes are **concepts** (words / phrases);
- edges are **named relations**.

Typical relations: `IsA`, `AtLocation`, `UsedFor`, `CapableOf`, `Causes`, `PartOf`, `HasProperty`.

Remarks

Common sense *written down* as a graph:
millions of obvious facts, each a typed arrow
between two ideas.



Other triples: `knife-UsedFor->cutting`,
`bird-CapableOf->fly`.

🍷 The benchmark paper: Talmor et al. (2019) ---

From here until **the gap** (the large **model–human accuracy gap**), the Background follows **one paper: Talmor, Herzig, Lourie & Berant (2019) — CommonsenseQA** (NAACL-HLT).

- **Purpose:** build a test that **forces common sense** — a model passes **only** by using **everyday world knowledge it already has**.
 - ▶ **No passage to look up** → the answer is *not* written in the text.
 - ▶ **No word-matching shortcut** → you cannot win by picking the option that just “sounds related”.
- **Key idea:** build every question from **ConceptNet** so all answer choices are **close siblings (same source concept, same relation)**.
 - ▶ They all “sound related” **equally** → word-matching **fails**.
 - ▶ **Only reading the specific situation** picks the

Where it sits

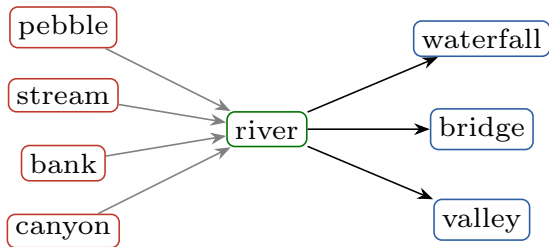
ConceptNet (Speer 2017) is the *ingredient*; **CommonsenseQA** (Talmor 2019) is the *benchmark* built from it. The two focal papers come *after*.

Example — why matching fails

From **river** (AtLocation): **waterfall, bridge, valley** — all “sound related” to *river* **equally**. Only the **specific situation** (“hold a cup upright to catch water”) points to **waterfall**.

🍷 From ConceptNet to questions: how CQA is built ---

a) Sample ConceptNet for specific subgraphs



b) Crowd-source questions + two distractors

Where on a **river** can you hold a cup upright to catch water on a sunny day?

✓ **waterfall**, **bridge**, **valley**, **pebble**, **mountain**

Where can I stand on a **river** to see water falling without getting wet?

waterfall, ✓ **bridge**, **valley**, **stream**, **bottom**

I'm crossing the **river**, my feet are wet but my body is dry, where am I?

waterfall, **bridge**, ✓ **valley**, **bank**, **island**

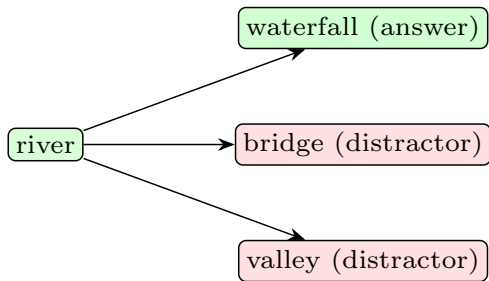
Talmor et al. (2019), Fig. 1.

An **Amazon Mechanical Turk (MTurk)** crowdworker **HG1**, the question-writers — paid humans who *write data* — see a **source concept** (*river*, green) and **three target concepts** (*waterfall*, *bridge*, *valley*, blue) sharing one relation (*AtLocation*). **HG1**, the question-writers (Talmor et al. 2019) writes **three questions**, one per target as the answer; the other two are distractors.

Then per question: **+1 ConceptNet distractor** (red) and **+1 hand-authored** (purple) → **5 choices** total.

🍷 Why the distractors make it hard ---

The distractors are **ConceptNet siblings**:
same source concept, same relation.



So a model **cannot** win by spotting which option “sounds river-ish” — they all do. It must read the specific situation (*hold a cup upright to catch water*).

Key insight

The construction *forces* commonsense: surface word-association is neutralised by design, because every option is equally associated with the source.

The model-human gap

Humans (**HG2**)

89%

BERT-large (best model '19)

56%

Gap = 89% - 56% = **33 pts**

The Google-snippets test

Source: top results from a **web search engine** (Google), retrieved per question.

Selection: query the question; keep the **highest-ranked snippets**.

Given to: **BIDAF++**, a

and the gap ---

BERT as the evaluation baseline. An examiner who designs an assessment, then administers it to a cohort of candidates:

- 1 **Construction of the instrument.** Talmor's contribution is the benchmark itself (~12,247 CommonsenseQA items) — the *dataset*, not any model.
- 2 **Administration to candidate models.** The benchmark is administered to several 2019 systems; among them **BERT-large**, the rest weaker or of earlier design.
- 3 **Evaluation protocol.** Each model is fine-tuned on the training partition, then assessed on the held-out test partition (standard supervised evaluation).
- 4 **Comparison of results.** BERT-large attains the highest accuracy, yet only $\approx 56\%$, against $\approx 89\%$ for the **human respondents** (**HG2**, the test-takers) — not the writers.

Interpretation: if even the strongest contemporary model reaches only 56%, models do not yet exhibit genuine

🍷 Talmor et al. (2019): the conclusion ---

Data obtained (measured)

- Best model **BERT-large** $\approx 56\%$ (5-choice test).
- **Humans** (**HG2**, the test-takers) $\approx 89\%$ on the same test.
- Adding **retrieved web snippets**: no accuracy gain.

Scientific conclusions

- The model-human gap is **large** $\rightarrow$ today's models **do not genuinely reason** with commonsense.
- Since extra text doesn't help, the limit is **reasoning with world knowledge, not missing data**.
- By construction the benchmark **isolates commonsense** $\rightarrow$ a sound, shared yardstick.

Why it matters for us

The lecture's **shared backdrop**: **Paper 1 (Rajani)** is evaluated on it; the open question — *how* do these models reason? — launches the two focal papers.

🍷 Background → the two papers ---

The set-up for the lecture

Given this hard benchmark, two natural questions follow:

- **Paper 1 (Rajani et al., 2019):** can a model that *writes its reasoning* answer better? → *generate* explanations.
- **Paper 2 (Yin & Neubig, 2022):** can we *point at the exact input word* that drove a prediction? → *localise* evidence.

verbalise the reasoning

vs.

localise the evidence

The two papers at a glance ---

Unit IV · Week 12. Both methods explain a trained **language model (LM)** *from the outside* (inputs → outputs); the LM differs per paper. *Two key terms:* **plausibility** = looks convincing to a person; **faithfulness** (= fidelity) = reflects the model's *true* computation.

Paper 1 · Rajani (2019) — CAGE

CAGE = Commonsense Auto-Generated Explanations.

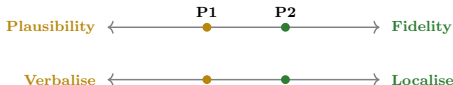
Verbalise the reasoning

- **Explains:** a BERT classifier (few options).
- **What:** a natural-language narrative.
- **Goal:** plausibility + accuracy.
- **Family:** generated explanation.
- **Nature:** black-box (internals not seen).

Paper 2 · Yin & Neubig (2022)

Localise the evidence

- **Explains:** a generative LM — GPT-2 / GPT-Neo (~50k tokens).
- **What:** attributes output to input tokens.
- **Goal:** faithfulness, fine-grained evidence.
- **Family:** contrastive attribution (grad./erasure).
- **Nature:** gradients (white-box) or perturbation.



Explain Yourself! Leveraging Language Models for Commonsense Reasoning

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Proceedings of the 57th Annual Meeting of the ACL, 2019 (arXiv:1906.02361)

🍷 What is CAGE? ---

FRAMING

CAGE = Commonsense Auto-Generated Explanations.

A framework that uses a **language model** to *automatically generate* a natural-language **explanation** for each question, then feeds that explanation as **extra input** to a classifier that picks the answer.

- **Origin / authorship:** introduced by **Rajani, McCann, Xiong & Socher (Salesforce Research)** in *Explain Yourself!*, **ACL 2019**.
- **Why it mattered:** the first method to *generate* reasoning text and use it to **improve** commonsense QA — it raised the **state of the art on CommonsenseQA v1.0** by $\approx 10\%$.

A note on terminology

Two terms recur and must be kept distinct. **CoS-E** names the *resource* — the dataset of human explanations (free-form rationales and highlighted spans). **CAGE** names the *method* — a language model, fine-tuned on CoS-E, that *produces* the rationale subsequently supplied to the answer classifier.

The acronym's expansion is not standardised: the paper's title uses the singular (*Commonsense Auto-Generated Explanation*), whereas the literature favours the plural (*Explanations*), adopted here.

Convention for these slides: **CoS-E** always denotes the data, **CAGE** the model.

Introduction + Motivation ---

FRAMING

The problem: deep models do poorly on tasks needing commonsense reasoning — knowledge not present in the input.

- An **explanation** verbalises the reasoning a model uses.
- **CommonsenseQA** is the benchmark (Talmor et al., 2019) — see Background.
- Open question: *do* these models reason, and how much rests on world knowledge?

Core idea

Train a language model to **generate explanations**, feed them to a classifier, and **measure whether they help**.

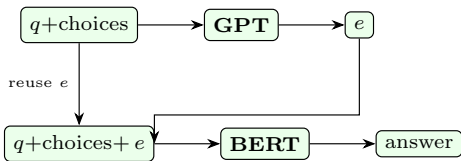
Remarks

Test whether the explanation *helps accuracy* — not merely whether it sounds nice.

🍷 The pipeline, end to end ---

- 1 Take a CQA example: question q , choices c_0, c_1, c_2 , gold answer a (the ground-truth answer).
- 2 A **different** crowd of humans (**HG3**, the explanation-writers; MTurk, *not* **HG1**) writes explanation e_h for *why* a is correct (this is **CoS-E** — Common Sense Explanations).
- 3 Fine-tune a language model (**GPT** — generative pre-training, Radford et al., 2018) to generate $e \approx e_h$.
- 4 Concatenate $q+\text{choices}+e$ and feed to a **BERT** classifier (bidirectional encoder, Devlin et al., 2019).
- 5 Measure: does e raise accuracy?

METHOD — OVERVIEW



Remarks

GPT explains, BERT decides. The LM is a *commentator*, not the *judge* — roles unpacked next slide.



Two models, two roles: GPT narrates, BERT rules ---

METHOD — OVERVIEW

A **legal metaphor** for CAGE's two-model design:

- **GPT — the commentator.** The *generator*. Given q +choices it **writes** the explanation e ; it never selects the answer. It narrates *why* an option could make sense.
- **BERT — the judge.** The *classifier*. It reads q +choices+ e and **delivers the verdict** — the predicted answer that is actually scored.

A commentator can give a compelling account, but only the judge's ruling counts. This is exactly why CAGE measures **plausibility**, not fidelity: the narrator's story need not be what the judge truly used to

The two architectures of the era (2018–19)

OpenAI GPT (GPT-1; Radford et al. 2018): a *decoder-only* Transformer (Vaswani et al., 2017), *left-to-right* (autoregressive), ~117M params, pre-trained on BooksCorpus. Built to **generate** text.

BERT (Devlin et al. 2019): an *encoder-only* Transformer, *bidirectional* (masked-language-model pre-training); fine-tuned for multiple choice by a classifier on the [CLS] (classification) token. Base 110M / large 340M. Built to **understand / score**.

🍷 Didactic aside: what does "fine-tune GPT to generate e" mean? ---

METHOD — OVERVIEW

- A language model already knows how to **continue text** (next-word prediction).
- **Fine-tuning** = keep training it, but now on (question + choices) -> **explanation** pairs from CoS-E.
- After fine-tuning, given a *new* question it can **write its own explanation**, imitating the human ones.

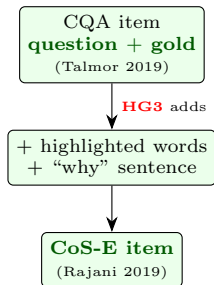
Mental model

We are not teaching new facts; we are teaching a *format*: "given a question, produce the kind of one-sentence justification a person would write."

🍷 How CoS-E is built from CQA - step by step ---

DATASET — CoS-E (HUMANS)

- 1 **Start from CQA.** Take CommonsenseQA (Talmor et al. 2019) — the questions **HG1** built from ConceptNet.
- 2 **Pick one item.** A question together with its correct (gold) answer.
- 3 **Give it to HG3.** A **different** MTurk crowd reads that question + gold answer.
- 4 **Highlight.** They mark the words that justify the answer → **CoS-E-selected**.
- 5 **Explain.** They write **one sentence** saying *why* the answer is right → **CoS-E-open-ended**.
- 6 **Filter & collect.** Drop bad ones; now every CQA item carries a human explanation = **CoS-E**.



Sizes (train / dev): v1.0 = 7,610 / 950;
v1.11 = 9,741 / 1,221.

Why it matters

CoS-E is the **supervision** for **CAGE** (next): the **only** thing humans

🍷 CoS-E: three concrete examples (Table 1) ---

DATASET — CoS-E (HUMANS)

From the paper

Q: “While eating a hamburger with friends, what are people trying to do?” Choices: **have fun**, tasty, indigestion.

CoS-E: “Usually a hamburger with friends indicates a good time.”

Q: “After getting drunk people couldn't understand him, it was because of his what?” Choices: lower standards, **slurred speech**, falling down.

CoS-E: “People who are drunk have difficulty speaking.”

Q: “People do what during their time off from work?” Choices: **take trips**, brow shorter, become hysterical.

CoS-E: “People usually do something relaxing, such as taking trips, when they don't need to work.”

🍷 CoS-E: a worked example, link by link ---

DATASET — CoS-E (HUMANS)

Question

While eating a hamburger with friends, what are people trying to do? — **have fun** / tasty / indigestion.

The bare question never says friends are enjoyable.
The explanation supplies the missing world-knowledge link:

hamburger + friends → social → enjoyable →
have fun

Remarks

That chain — “hamburger with friends = a good time” — is commonsense made *explicit in words*. That is what CoS-E records.

🍷 What is "CSRM" - and why is BERT the one doing it? ---

METHOD — CAGE (ALGORITHM)

- 1 **CSRM = Commonsense Reasoning Model.** It is a **role**: “the model that decides the answer”.
- 2 That role is a **classification** job — read q + choices + explanation, then **pick one** option.
- 3 They fill the role with **BERT**, because BERT is a **classifier** (built to read and score). GPT *cannot* do this — it only **generates** text.
- 4 So “**CSRM (BERT)**” just names **who fills the role** — like writing “the judge (Dr. Pérez)”.

Why BERT *can* pick an answer

BERT is **fine-tuned** for multiple choice by adding a tiny classifier on its special [CLS] token — that is what lets it **choose** one of the options.

Memory hook

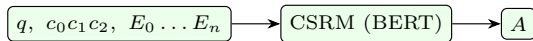
GPT narrates, BERT judges. The narrator writes the story; only the judge's ruling counts.

🍷 The two phases: GPT writes the "why", BERT picks the answer ---

(a) generate — GPT writes the explanation



(b) classify — BERT picks the answer



Reading the symbols

q = the question; $c_0 c_1 c_2$ = the 3 answer choices.
 $E_0 \dots E_n$ = the explanation, written **token by token** (a **token** $\approx$ a word).
 E_i = the **next** word being written; A = the chosen answer.

METHOD — CAGE (ALGORITHM)

- 1 **Phase (a) — generate.** GPT reads q + choices + **the words written so far** ($E_0 \dots E_{i-1}$) and writes **the next word** E_i . Repeat until the explanation is finished.
- 2 **Phase (b) — classify.** The other model reads q + choices + **the finished explanation** ($E_0 \dots E_n$) and picks the answer A .

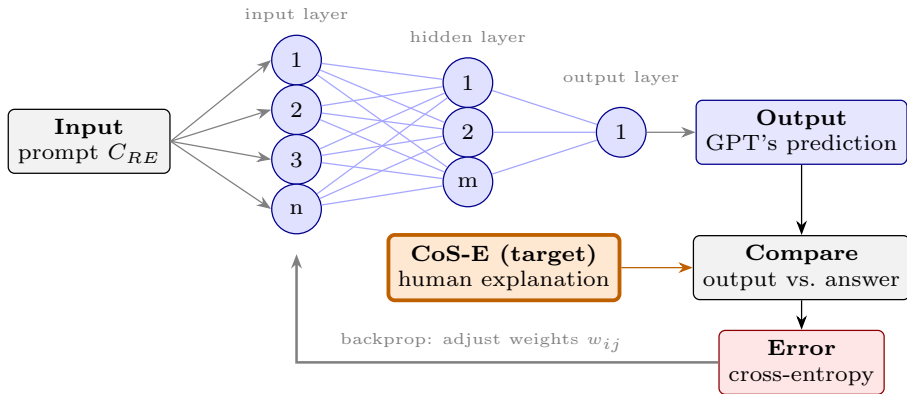
Half-written vs. finished

In (a) the explanation is **half-written** (up to E_{i-1}); in (b) it is **complete** (up to E_n).



How GPT learns to explain - the training loop ---

METHOD — CAGE (ALGORITHM)



Read it: the prompt enters the network; the prediction is **compared** with the human explanation (**CoS-E**, the target); the **error** flows back (backprop) to adjust the weights. CoS-E never enters as input — it is the **reference for correction**.



Where the prompts come from - and what CoS-E is ---

METHOD — CAGE (ALGORITHM)

Where do the prompts come from? C_{RE} is **not a stored dataset** — it is the **template** (the “mould”) of the reasoning prompt. There is **one C_{RE} per CQA question** (one per item), **assembled on the fly** from $q+$ choices + a fixed template. No prompt is written by hand.

Example prompt — reasoning (C_{RE}): answer *absent*

“...have fun, tasty, or indigestion? *commonsense* says _____”

Example prompt — rationalisation (C_{RA}): answer *present*

“...have fun, tasty, or indigestion? **have fun** *because* _____”

Example CoS-E explanation — the *target* GPT must learn

“a hamburger with friends usually means a good time”

The whole game: is the answer in the prompt or not? C_{RE} **reasons** (before the answer);
 C_{RA} **justifies** after the fact.

🍌 Results: CommonsenseQA v1.0 (3 choices) ---

EVALUATION — RESULTS

Dev (random split)

Method	Acc. (%)
BERT baseline	63.8
CoS-E-open-ended	65.5
CAGE-reasoning	72.6

Test split (v1.0)

Method	Acc. (%)
RC (Talmor)	47.7
GPT (Talmor)	54.8
CoS-E-open-ended	60.2
CAGE-reasoning	64.7
Human (HG2)	95.3

Reading the tables — what each term means

Accuracy (%) — the share of multiple-choice questions answered correctly; the sole metric. **Dev vs. Test split** — the data is partitioned: the *development* split is used while tuning, the *test* split is held out for the final score. **Baseline** — a reference system without the proposed mechanism (BERT, no explanations). **Human (HG2)** — the *human ceiling*: accuracy attained by people. **RC** = reading-comprehension baseline (Talmor et al., 2019).

Headline

CAGE-reasoning gains ≈ 10 percentage points (**absolute**, i.e. $64.7 - 54.8$) over the previous **state of the art** on test — yet stays far below the **human ceiling** of 95.3%.

🍷 Results: CQA v1.11 (5 choices) – the uncomfortable result ---

EVALUATION — RESULTS

What is v1.11? The **harder, 5-choice** edition of CommonsenseQA (v1.0 has 3 choices). The *uncomfortable result*: here the proposed method **underperforms** the plain baseline.

Method	Acc. (%)
CAGE-reasoning	55.7
BERT baseline	56.7
CoS-E-open-ended	58.2

Remarks

On the harder 5-choice version **CAGE loses to the plain baseline** (55.7 vs. 56.7). The generated explanation often *contains* the correct answer, yet the classifier cannot exploit it.

Connect to background — why it is hard

ConceptNet siblings: in v1.11 the five answer choices are all drawn from *related* ConceptNet nodes, so they are **semantically very close**. With options that similar, merely *concatenating* the explanation is not discriminative enough. The authors report this negative result **honestly**.

🍷 Results: out-of-domain transfer ---

EVALUATION — RESULTS

What is out-of-domain transfer? Applying the explanation module **trained on CQA** to **different tasks**, with **no retraining**, to test whether the explanations still help.

Method	SWAG	Story Cloze
BERT	84.2	89.8
+ explanation transfer	83.6	89.5

Remarks

Transfer without retraining costs only a **tiny drop** ($< 0.6\%$). The explanations are fluent and relevant, but give **no downstream gain** (*downstream* = on the end task). An *honest negative result*.

Datasets — the table columns

SWAG = Situations With Adversarial Generations (Zellers et al., 2018) — commonsense inference about everyday situations. **Story Cloze** (Mostafazadeh et al., 2016) — choosing the correct ending of a short story. Both are **separate** tasks from CommonsenseQA.

🍷 Qualitative analysis - what the explanations look like ---

ANALYSIS — LIMITATIONS

- **Form:** CAGE uses **simpler constructions** than the humans' (**HG3**), yet can be *more* informative.
- **Content:** an answer choice appears **43%** of the time, but the *predicted* choice only **21%** → the explanation is **not a circular echo** of the chosen answer.
- **Similarity to humans:** BLEU peaks at **4.1** (vs 0.8 untuned) — **very low**. **Fluency:** perplexity **32** — low, i.e. fluent.

Foreshadowing

Low BLEU but real usefulness → something can be *useful without resembling human wording*. This previews Paper 2's **faithfulness vs. plausibility** tension.

The two metrics, defined

BLEU (Papineni et al., 2002) — measures **how closely a generated text matches human reference texts** by counting shared word sequences (n-grams). Range 0–100; **higher = more similar** to the references. The value 4.1 means CAGE's wording is **far** from the humans'.

Perplexity — measures **how fluent / predictable** a text is for a language model; formally the exponential of the average per-word cross-entropy. **Lower = the model is less “surprised” = more fluent text**. The value 32 indicates fluent output.

🍷 Limitations + critical reading of Rajani et al. ---

ANALYSIS — LIMITATIONS

- **Human simulatability is low:** from the explanation alone, **HG4a**, Rajani's Turkers (evaluators who *judge*, not writers) recover the model's answer **42%** (CAGE) vs **52%** (human) → the explanation does not transparently reveal the model.
- **Adversarial explanations are catastrophic:** misleading explanations drop accuracy **60%** → **30%** — below the 50% baseline.
- **Bias propagation:** CQA gender disparity flows into CoS-E and the trained models.

What Rajani et al. leave open

They *generate* explanations, but never ask:
which input tokens caused this prediction,
and **why this token instead of another?**

- Generated explanations are **plausible** but not necessarily **faithful**.
- Language generation has an enormous output space → we need a **token-level** lens.

Next

Yin & Neubig (2022): contrastive input-*saliency* (which input tokens mattered) — look *inside* the model.

Interpreting Language Models with Contrastive Explanations

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Proceedings of EMNLP 2022 (Outstanding Paper Award) (arXiv:2202.10419)

Paper 2 in a nutshell: what it is about ---

Before the details, here is the whole paper in eleven short steps. The question it answers is deceptively simple: *why did the model choose this word and not another?*

- 1 Classic interpretability methods work well for **classification** (*cat* vs *dog*), but not for language models, which choose among thousands of possible words.
- 2 A single word can be chosen for many reasons **at once**: meaning, tense, number, syntax, style.
- 3 Asking plainly “*why this word?*” tends to produce **confusing** explanations.
- 4 The authors pose a different question: “*why this word **rather than** that one?*”.
- 5 They call this a **contrastive explanation**.
- 6 Instead of explaining everything, it explains **only the difference** between two candidates.
- 7 Example: explain why the model wrote *themselves* and not *herself*.
- 8 The method identifies **which parts of the context** favour one option over the other.
- 9 This surfaces the semantic and grammatical factors in a form that is far **clearer to humans**.
- 10 The resulting explanations are **more compact and more interpretable** than traditional saliency.
- 11 The central idea: **to understand a decision is to understand the boundary that separates one alternative from another** — not to explain the whole space of possibilities.

🍷 Paper 2 in plain words: the contrastive idea ---

The problem. A language model picks the next word among $\sim 50,000$ options. Asking “*why this word?*” gives a vague answer — it just points at the word right before, and blurs grammar, number and meaning together. We cannot see what really drove the choice.

The fix: ask a contrastive question

Instead of “*why did it say X?*”, ask “*why X rather than Y?*” — comparing against a rival turns a vague question into a precise one.

Worked example

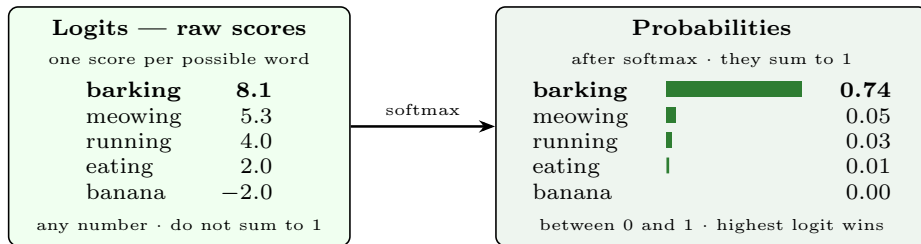
Input: “Many teenagers were helping _____”. The model says **themselves** (not *herself*).

Why “themselves” rather than “herself”? → because of the word “**teenagers**” — it is *plural*, so it requires *themselves*. The plural subject is what tipped the decision.

Change the rival, change the question: the rival (*Y*) aims the explanation at one specific decision.

Under the hood: logits and the softmax ---

Before the subtraction itself, three pieces of machinery must be made explicit. The first is how a language model turns its internal computation into a single chosen word.

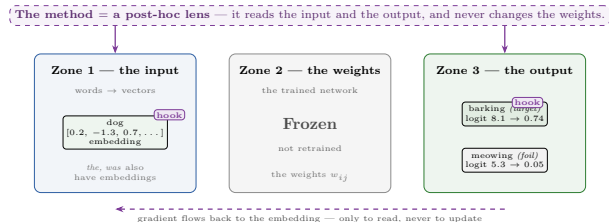


Logits and the softmax, defined

A language model holds a fixed **vocabulary** (about 50,000 word-pieces for GPT-2). At each step it assigns every word in that vocabulary a raw, unnormalised score, called a **logit**; these scores may be positive or negative and do not sum to one. The **softmax** function then maps the full set of logits to a **probability distribution** — values between 0 and 1 that sum to 1. The word with the largest logit receives the largest probability and is selected as the output.

🍲 Under the hood: the model in three zones ---

The second piece is a map of the model in three zones. It makes precise *where* the method acts and, just as importantly, where it does not.

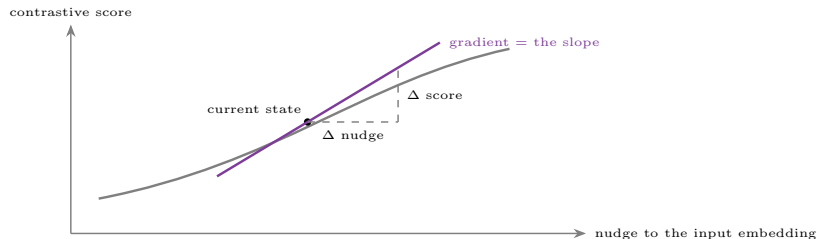


Three terms, defined

Embedding — the numeric vector into which each input word is converted before the network reads it. **Logit** — the raw output score of a candidate word, before the softmax. **Hook** — an instrumentation point that reads an internal value (or its gradient) without altering the computation. The method is **post-hoc**: the weights of Zone 2 stay frozen.

🍷 Under the hood: the gradient as a sensitivity ---

The third piece is how a word's influence is actually measured. It is read from a **gradient** of the contrastive score with respect to the input embedding.



A rate, not a difference

The gradient is a **rate**: the change in the contrastive score *per unit* change in the input embedding — a slope, not the change itself. The two gradient methods use this slope (one cheap backward pass). The **erasure** method instead measures a **finite difference**: it deletes the word, re-runs the model, and reads how much the score actually moved. **Gradients approximate the perturbation; erasure performs it.**

The flagship example: "barking", not "crying" ---

The paper's opening example (Table 1). The model reads "*Can you stop the dog from ____*" and predicts **barking**. Each row colours the input: **red** raises the probability of the target, **blue** lowers it, white means little influence.

(1) why <i>barking</i> ?	Can	you	stop	the	dog	from
(2) <i>barking</i> vs <i>crying</i>	Can	you	stop	the	dog	from
(3) <i>barking</i> vs <i>walking</i>	Can	you	stop	the	dog	from

Reading the three rows

(1) **Plain saliency is uninformative**: almost all the weight lands on the preceding token "from". (2) **Against "crying"**, the word "**dog**" lights up — dogs bark, they do not cry. (3) **Against "walking"**, the word "**stop**" lights up — one stops an *unwanted* behaviour. **Change the foil, change the evidence.** (In the paper's other example, "Spanish" vs "Portuguese" makes the word "Spain" salient.)

The three methods, in plain words ---

All three answer the *same* question — **how much does this word push towards the target rather than the foil?** — and rank the words by that push. They differ only in the **instrument** used to read it.

Formally (reference): contrastive slope $g^* = \nabla_{x_i} (q(y_t|x) - q(y_f|x))$; $S_{GN}^* = \|g^*\|_{L1}$, $S_{GI}^* = g^* \cdot x_i$, erasure deletes x_i and re-measures.

The three instruments

1 – Gradient Norm (“does it matter?”): the **size** of the slope, no direction. Cheap; needs the model’s internals.

2 – Gradient $\times$ Input (“towards which option?”): the same slope, now **signed** (+ target, – foil). Cheap, and more informative than the norm.

3 – Input Erasure (“delete it and see”): no slope — **remove the word, re-run, measure the real change**. Direct and needs no internals, but **expensive** (one run per word).

In one line

Gradient $\times$ Input is the practical winner (cheap, signed, best or near-best everywhere);

Erasure is the most faithful but costly; the **Norm** is the simplest but says the least. (Gradients *approximate* the push; erasure *performs* it.)

Grading an explanation objectively: BLiMP ---

BLiMP (Warstadt et al., 2020) provides *minimal pairs*: two sentences differing in one spot, one grammatical and one not. The token that enforces the rule is the **known evidence**, fixed in advance.

Acceptable: “Many teenagers were helping **themselves**.”

Unacceptable: “Many teenagers were helping **herself**.”

known evidence:
the antecedent
“**teenagers**” (plural)

The setup

5 phenomena / 12 paradigms (anaphor, argument structure, determiner–noun, NPI, subject–verb). Models: **GPT-2** (1.5B), **GPT-Neo** (2.7B). The test: does the **top-ranked token** land on the known evidence?

The alignment metrics ($\mathcal{E}$: evidence, $\mathcal{S}$: saliency)

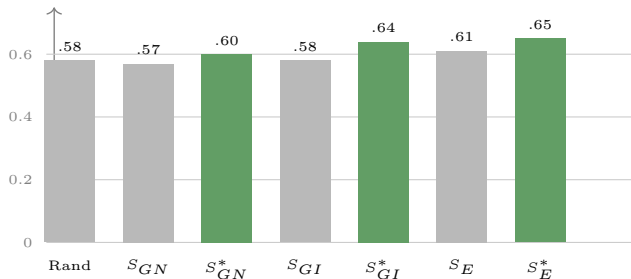
Dot product $\mathcal{S} \cdot \mathcal{E}$ (**higher** better); **Probes needed** — rank of the first evidence token (**lower** better); **MRR** — mean of $1/\text{rank}$ (**higher** better).

Why this matters: from “plausible” to “gradeable”

BLiMP turns interpretability into a **gradeable exam**. Without it, the most one can say is “*my explanation looks reasonable*”; with it we can ask “*did the method land on exactly the word grammar says should matter?*” That shift — **plausible** → **checkable** — is what lets Yin & Neubig **prove** contrastive explanations beat traditional saliency.

🍷 RQ1 - contrastive explanations align better ---

RQ1: do contrastive explanations find the linguistically appropriate evidence? Mean alignment with the known BLiMP evidence (here MRR, GPT-2; *higher is better*). Contrastive variants S^* in **green**.



What the bars say

For both GPT-2 and GPT-Neo, and across all three metrics, the **contrastive** variant beats its non-contrastive twin; the non-contrastive

The distance effect

The **further** the evidence token sits from the prediction, the **larger** the contrastive advantage — contrastive explanations

🍷 RQ2 - contrastive explanations help humans ---

RQ2: do the explanations help a person predict the model? (simulatability — shown the input + the explanation, can a person guess the model's next word?). **10 ML graduate students, 4,000** judgments; the model is right exactly 50% of the time, so guessing the grammatical option does not help. **Higher is always better.**

Method	Acc.	Corr.	Inc.	Use.
None	61.38	74.50	48.25	—
S_{GI}	64.00	78.25	49.75	62.12
S_{GI}^*	65.62	79.00	52.25	63.88
S_E	63.12	79.00	47.25	46.50
S_E^*	64.62	77.00	52.25	64.88

Rows. **None** = no explanation shown; S = non-contrastive, S^* = contrastive (the *); GI = gradient $\times$ input, E = erasure.

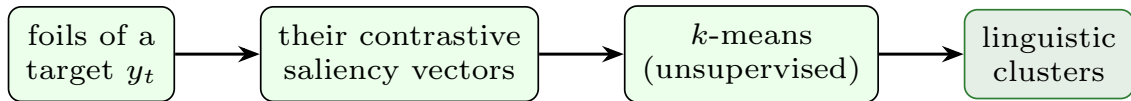
Columns. **Acc.** = % of times the user guessed the model's word; **Corr./Inc.** = the same, on cases where the model itself was right / wrong; **Use.** = % of explanations users called helpful.

Conclusions, read straight from the numbers:

- ➊ **Any explanation helps.** With none, users hit 61.38; every method climbs to $\geq 63 \Rightarrow$ explanations make the model more predictable.
- ➋ **Contrastive beats plain.** grad $\times$ input 64.00 $\rightarrow$ **65.62**; erasure 63.12 $\rightarrow$ **64.62**. The starred (S^*) version wins in *both* families.
- ➌ **It helps most where intuition fails.** On *Incorrect* cases (model $\neq$ grammatical guess): 48.25 (None) $\rightarrow$ **52.25** (both S^*) — the largest jump.
- ➍ **Users feel it too.** “Useful” rises with contrast: grad $\times$ input 62.12 $\rightarrow$ **63.88**; erasure leaps 46.50 $\rightarrow$ **64.88**.
- ➎ **Best overall:** S_{GI}^* (**65.62**) — cheap and most accurate; S_E^* is close and rated the most useful.

RQ3 - clustering decisions by their cause ---

RQ3: do different decisions need different evidence, and does that evidence reveal coherent linguistic concepts? Represent each **foil** by the saliency vector it produces against a target, then cluster those vectors.



Clusters recovered without supervision

A **male** pronoun target $\rightarrow$ a cluster of **female** pronouns; an **animate** noun $\rightarrow$ a cluster of **inanimate** nouns; a **singular** noun $\rightarrow$ a cluster of **plural** foils. These differ from word-embedding neighbours — they reflect **what the model uses to decide**.

Which evidence for which decision

Gender (pronouns, determiners): gendered proper nouns (“Veronica”, “he”).

Number/verbs: other verb forms.

Adjectives: semantically related words (to tell “black” from colours, “relativity” matters). **Numbers**: enumeration words (“age”, “least”).

Limitations, and what to take away ---

Where the method is fragile, and what it contributes.

Limitations

- **Foil selection** is consequential: open-ended generation has no rule for which foil to use, and changing it changes the result.
- **Plausibility** $\neq$ **faithfulness**: matching human intuition does not prove the saliency reflects the true computation.
- **Cost**: erasure is the most direct but scales poorly with length and foil-space size.

Contributions

- Extended **three** saliency methods to the contrastive setting (norm, $\text{grad} \times \text{input}$, erasure).
- Proposed **three** evaluations: alignment (BLiMP), human simulatability, and clustering-based analysis.
- Contrastive explanations give a **more intuitive, fine-grained** interpretation of LMs.

The thread back to Paper 1

Rajani **verbalises** reasoning (optimised for **plausibility**); Yin & Neubig **localise** evidence (aimed at **faithfulness**). Two ends of the same question: *how do language models reason?*

🍷 Recap - the two papers at a glance ---

Recap. Both methods explain a trained **language model (LM)** *from the outside* (inputs $\rightarrow$ outputs) — but the LM itself differs per paper.

Paper 1 · Rajani et al. (2019) — CAGE

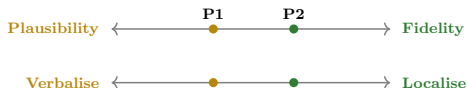
Verbalise the reasoning

- **Explains:** a BERT classifier (few answer options).
- **What:** a natural-language narrative.
- **Goal:** plausibility + accuracy.
- **Family:** generated explanation.
- **Nature:** black-box (internals not inspected).

Paper 2 · Yin & Neubig (2022)

Localise the evidence

- **Explains:** a generative LM — GPT-2 / GPT-Neo ($\sim 50,000$ tokens).
- **What:** attributes the output to input tokens.
- **Goal:** faithfulness, fine-grained evidence.
- **Family:** contrastive attribution (gradient/erasure).
- **Nature:** gradients (white-box) or perturbation.



Open questions ---

- Can generated explanations (Paper 1) be **constrained to be faithful** via contrastive saliency (Paper 2)?
- What is the right way to **choose foils** for open-ended generation?
- For regulated settings (EU AI Act — the EU's AI regulation, Art. 13), is a **plausible** explanation enough, or is **faithfulness** legally required?

Discussion?

🍷 Thank you ---

Thank You!



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